GPUOS: An Operating-System Architecture for Shared GPU Execution

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***Abstract*—Latency-sensitive inference can leave arithmetic capacity idle while occupying resources needed by other GPU work. Recovering that capacity requires a contract for interference, suspension, and resource return. Building on *The GPU Needs an Operating System*, we specify GPUOS, an architecture that represents execution as restartable tasks admitted under revocable resource leases. Its six layers connect compiler safe points, device dispatch, resource governance, interference estimates, and slack-aware admission. The design separates cooperative suspension, finite kernel slices, and driver-managed context switching, and identifies hardware services needed for enforceable pause and memory-quality guarantees. Component prototypes provide evidence on one 32-GB RTX 5090: bounded fused inference, a restricted FX-to-Triton compiler, cooperative background work, and cross-process admission. Qualified fused search reduces complete-request latency by 17.7% and 18.2% against native cached execution, while a matched CLC+PDL control remains faster. A streaming proxy admits 1.167 TFLOP/s of background work at 2.50% foreground p99 cost, but whole-step suspension recovers little work beside full-model replay. These measurements support selected mechanisms rather than a completed general-purpose implementation. The paper presents a unified system specification, its implemented subset, and the remaining compiler, service, and hardware requirements.**

I. Introduction

The useful unit of GPU ownership is rarely an entire device for the entire lifetime of an application. A model server needs a prompt response at some moments, a large KV allocation at others, and sustained memory throughput during decode. An adapter update or an evaluation job may tolerate interruption while consuming the arithmetic capacity that the server leaves available. Assigning the same indivisible device to either application hides these differences. Sharing the device without controlling interference can be worse: a throughput improvement is not useful if an interactive request misses its response-time objective.

Nourai and Bastani's *The GPU Needs an Operating System* makes this case through large-model measurements and a hardware–software research agenda [1]. We adopt its central organizing idea: GPU resources should be granted according to the urgent workload's remaining slack, rather than treating low average utilization as unconditional permission to run more work. We develop that idea into an architectural specification for GPUOS, covering process state, resource contracts, compiler checkpoints, admission, revocation, memory leases, and the hardware interface. This is a system-design paper supported by component experiments. The architecture includes services that the current prototype does not yet implement; the implementation and evaluation identify that boundary explicitly.

The resulting abstraction is a *resource lease attached to restartable work*. A lease specifies which resources a task may consume, how its progress is committed, and how much revocation delay admission must reserve. A task can run in a fused device executor, as a sequence of compiler-generated kernel slices, or in a separate protected CUDA context. These are execution backends with different costs and guarantees, not interchangeable meanings of preemption. The architecture chooses among them using a workload-specific cost model and preserves one accounting interface across the alternatives.

The prototype results expose limits as well as opportunities. Qualified fused search improves on native cached execution, but matched CLC+PDL remains faster. Cooperative sharing benefits a streaming proxy, while whole-step suspension yields little work beside full-model replay. The OS must therefore be able to reject an elaborate scheduling path when ordinary co-running is better.

Our contribution is an explicit architectural specification, an execution contract separating checkpoint safety from resource return, and component evidence for bounded implementations. The resource-vector and slack-driven sharing ideas originate in the cited work; new hardware services and a completed general-purpose implementation are not claimed.

II. Why GPU Sharing Needs an OS Contract

A. Decode and the roofline

Consider a matrix with W weights, represented by s bytes per weight, and a decode batch of B tokens. A weight-only approximation performs $2BW$ floating-point operations while reading sW bytes. Its arithmetic intensity is therefore

$$I_{\text{weights}} = \frac{2B}{s} \quad \text{FLOP/byte.} \tag{1}$$

For two-byte weights this is B; for one-byte weights it is $2B$. KV reads, activation traffic, routing, and synchronization add costs that the expression does not capture. The device ridge $R = F_{\text{peak}}/\beta_{\text{mem}}$ compares peak arithmetic throughput and memory bandwidth at a consistent precision. A memory-bound operation has an idealized arithmetic-utilization bound $\min(1, I/R)$. A low value is a structural property of the operation, not evidence that a dispatch queue was implemented poorly.

For a mixture-of-experts layer, the relevant batch is the batch reaching one expert. Under balanced routing, E experts and k active experts per token give an average $B_e = Bk/E$. A large global batch can consequently remain a small local matrix problem. Routing imbalance makes the average a poor

description of the tail. This matters for an OS: scheduling against the global token count can overestimate the arithmetic reuse of each ready task, while the latest-finishing expert determines when dependent work is released.

a) Evidence motivating the design.: OpenInfer reports roughly 7% tensor activity for Qwen3.5-397B at a 97.2% deadline-attainment operating point on four RTX PRO 6000 cards, and 2.9% in its decode-only pool [1]. Its Kimi K2 experiments report 6–11% activity at carried loads. These are that paper's serving measurements, not our RTX 5090 observations. Its streaming control reaches approximately 99% of measured bandwidth on 24 of 188 SMs, whereas its foreground reaches only about half of peak bandwidth. This contrast motivates jointly choosing a foreground kernel and its allocation.

B. Why a scheduler alone is insufficient

Batching can increase reuse, but waiting for a larger batch consumes response time and larger steps can increase token latency. A scheduler cannot remove that tradeoff by rearranging identical non-preemptible kernels. It needs additional work that is both useful and compatible with the urgent phase. Compatibility depends on more than arithmetic intensity: a cache-resident matrix loop and an operand-streaming GEMM may issue similar tensor work while causing very different foreground interference.

Nor can the scheduler infer available SMs from low bandwidth utilization. A poorly sized foreground kernel may need many SMs to generate enough outstanding memory requests. Taking those SMs away reduces the foreground's own throughput before the background transfers a single byte. Conversely, a well-pipelined streaming kernel may saturate bandwidth with fewer workers. The resource cost of a partition therefore has two parts: the foreground's loss from a smaller execution allocation and additional interference from the admitted background workload. The architecture measures these parts separately whenever a suitable matched control exists.

C. Time, space, and authority

CUDA stream priorities influence scheduling but do not promise to interrupt already-running work [2]. Green contexts constrain spatial execution resources; MPS coordinates multiple clients; MIG provides a different isolation boundary on supported devices [3], [4]. None should be described as a software-selected, instruction-level deadline primitive merely because it changes latency in one workload. Hardware context switching, compiler checkpoints, and static partitions address different parts of the problem.

We distinguish three authorities. An application can state urgency and declare its work. A runtime can admit or withhold future slices and manage buffers it owns. A driver or device can enforce context and physical-resource boundaries. The proposed OS API exposes the strongest available guarantee without silently upgrading it. In particular, a software issue-rate limit is not a memory-controller reservation, and a CTA launch limit is not a promise that a particular set of SMs belongs exclusively to that client.

III. System Architecture and Objects

A. A six-layer decomposition

We retain the source paper's six-layer decomposition as a design vocabulary [1]. L0 provides device signals, context scheduling, queues, and enforceable resource controls. L1 defines the obligations of a shareable kernel. L2 dispatches eligible work to execution resources. L3 governs resource consumption and memory capacity. L4 estimates interference and revocation costs. L5 selects admissions within foreground slack. Table I maps this specification to the current artifact.

B. Processes, tasks, and leases

A *GPU process* is the unit of ownership and accounting. Its identity names an address-space authority, loaded executable versions, memory allocations, and outstanding tasks. Trusted tasks may share one CUDA context and even one fused executor; mutually untrusted tasks require a protection boundary supplied below the runtime. The API makes this trust mode explicit rather than treating a common function-pointer table as process isolation.

A *task* names executable work and its dependency edges. Its descriptor contains a code version, tensor views, input/output access modes, a progress format, a resource declaration, and a deadline class. A *slice* is a bounded portion of that task with one well-defined commit boundary. A *lease* grants resources to one or more eligible slices for a bounded interval. A lease also specifies what must happen on revocation: retain a resident allocation, drain and return CTA resources, save to device memory, or invoke a driver-managed context transition.

C. Resource vectors and accounting

The proposed lease resource vector is

$$\mathbf{r} = (r_{\mathrm{SM}}, r_{\mathrm{TC}}, r_{\mathrm{L2}}, r_{\mathrm{DRAM}}, r_{\mathrm{cap}}, r_{\mathrm{link}}, r_{\mathrm{copy}}, r_{\mathrm{power}}). \quad (2)$$

The coordinates have different units. SM allocation concerns resident blocks, warp issue, registers, and shared memory. Tensor work concerns arithmetic issue. Cache and DRAM budgets concern traffic and queueing. Capacity is bytes held over time plus the cost of reclaiming them. Link and copy resources include direction and endpoint. Power affects clock rates shared by otherwise disjoint work. A scalar utilization percentage cannot substitute for this vector.

Declarations and measurements serve different purposes. A compiler can report the logical inputs and scratch required by a slice, along with the compiled register and shared-memory allocation. It cannot derive physical DRAM traffic exactly from those fields because cache hits, reuse, and other clients alter that traffic. The accounting service keeps both values, with explicit labels. If a counter is unavailable, it remains unknown; it is not filled with an arithmetic estimate and presented as a measurement.

Resource prices are foreground costs, not universal exchange rates. For foreground class f, background declaration $\mathbf{r}$, and execution state x, write $C_f(\mathbf{r}, x)$ for the additional completion time. The state includes phase, batch geometry,

Layer	Architectural responsibility	Artifact boundary
L0	Context switch, bounded pause, resource enforcement	Existing driver switching is measured; no new device pause or memory-QoS interface.
L1	Restartable work and declared resource footprint	Handwritten WMMA checkpoints and strict FX/Triton slices for supported tensor graphs.
L2	Ready-work selection and dispatch	Fused per-frontier device work pool, stream-ordered worker control, and a separate host slice executor.
L3	Capacity leases, issue pacing, cache/copy governance	Bounded arenas and CTA issue pacing; no physical cache or DRAM reservation.
L4	Workload-specific interference and revoke-cost estimates	Measured configuration lookup; no predictor for unobserved workloads.
L5	Deadline-aware admission and revocation	Held-out slack selection and one-foreground IPC admission; no integrated production service controller.

TABLE I
ARCHITECTURAL SPECIFICATION AND IMPLEMENTED EVIDENCE. THE TABLE IS A CAPABILITY MAP, NOT A CLAIM THAT ALL SIX LAYERS FORM A FINISHED DEPLOYED OS.

context length, device allocation, and recent interference. Even if individual resource effects are measurable, their combination need not be additive. Cache eviction can increase DRAM traffic, and a power change can alter the service time of every slice. The implemented lookup does not assume a linear price model; a more general predictor must be validated on resource combinations before admission uses it.

D. Events and executable versions

The architecture uses release/acquire publication between producers and consumers in the appropriate memory domain. A producer publishes the data and progress state before publishing completion. A consumer observes the completion event before reading those fields. A host-mapped doorbell requires system-scope ordering; a flag used only between kernels on the device can use device scope. Polling a volatile field without a publication protocol is not a replacement for these ordering rules.

Executable versions have lifetimes too. Hot-loaded code, device function pointers, and plan descriptors remain valid until every possible caller has retired. Replacing a jump-table entry does not prove that the old module is unreachable by a CTA that already loaded its pointer. The specification therefore associates a code epoch with each admitted task and reclaims an old version only after an epoch-completion condition. The existing fused executor uses precompiled handlers; general safe hot replacement remains an architectural service rather than a property established by these tests.

IV. COMPILER CONTRACT AND EXECUTION BACKENDS

A. Safe points are semantic boundaries

A compiler cannot make an arbitrary running GPU kernel safely restartable by inserting a flag read at an arbitrary instruction. Threads may disagree about which barrier phase they are in; shared memory may hold an unpublished partial result; or an external consumer may have observed an output that the resumed task would produce again. A checkpoint is legal only when its progress and externally visible effects define an unambiguous restart state. This requirement is stronger than limiting the number of loop iterations.

For a pure pointwise tile, a commit can consist of final stores to a disjoint output region. For a matrix product, a commit may consist of a partial accumulator and the completed K extent. A reduction similarly needs its partial value and reduction interval. The next invocation consumes exactly that state. The task's output becomes visible to dependent graph nodes only after all required tiles have committed. Intermediate numerical state may remain private even when some task-local work is complete.

The proposed general contract admits opaque kernels only if their provider supplies equivalent safe-point semantics or an acceptable non-preemptible bound. Unsupported side effects, device allocation, inter-block protocols, or external collectives cannot be assumed safe merely because the front end can capture their call site. Rejecting a graph is a valid compiler result. A silent eager fallback would violate the scheduler's assumed pause contract and is therefore inappropriate for this backend.

B. Implemented FX-to-Triton lowering

The current front end captures shape-specialized, pure inference graphs with FX and also exposes a `torch.compile` backend [5]. It lowers supported operations to Triton kernels and generated pointwise expressions [6]. It accepts floating CUDA tensors with checked shape, stride, dtype, and device specializations. Supported graphs include broadcast arithmetic, common activations, two-dimensional matrix products and linear layers, contiguous last-dimension sums and means, bounded-row softmax, and selected no-copy views. Autograd, mutation, unrecognized operations, and unsupported semantic options are rejected.

The compiler generates a sequence of stages rather than forcing every graph into a persistent megakernel. Each stage specifies its number of independent program instances, its reduction chunks, output storage, and any private partial storage. A matrix-product slice covers a bounded set of output tiles and a bounded K interval, accumulating in FP32. A sum or mean saves FP32 partials between reduction chunks. An elementwise stage maps a bounded range of logical elements through broadcasting and physical strides. A softmax row

currently executes as one program and is limited to 8192 elements.

The execution cursor identifies the stage, program offset, and reduction chunk. It lives in the host executor; partial arithmetic state lives in device allocations. The executor advances through all program ranges of a chunk before advancing the chunk, allowing successive grants to change their program counts without skipping or duplicating output tiles. The same stream orders those grants. Resuming on another stream requires an explicit handoff; the prototype rejects an unordered stream change.

Every slice is a finite kernel launch. Once that launch completes, its CTA resources are returned to the CUDA scheduler. This is the important semantic difference from a resident worker that sleeps. The cost is additional launch and synchronization work, more partial-state traffic, and possibly worse cache reuse. The compiler exports resource and progress metadata so that the controller can account for these costs. A maximum program count or K extent is a bound on work geometry, not a guaranteed number of microseconds.

C. Fused dynamic execution

The earlier fused executor, called FuseOS in the retained measurements, handles a different execution opportunity: changing work within a bounded model invocation. In output-driven search, scores determine the next live branch set, selected histories, and ready tiles. The model architecture alone does not fix those choices. Stable request identifiers and bounded storage plans separate changing logical ownership from physical reservations.

The evaluated Qwen3-0.6B path stores an immutable ancestor KV region and two private-history banks. Each frontier copies selected histories into the alternate bank, appends KV, and swaps banks. Precompiled handlers combine normalization and projection, gate/up and activation, and residual work. A cooperative persistent invocation executes all 28 layers and the vocabulary head for one frontier. Device work counters assign ready tiles within a phase; barriers order dependent phases. Host branch scoring remains part of the complete-request measurement. This is not a process-lifetime serving kernel with a GPU-owned external request queue.

The same arithmetic handlers support ordinary grids, programmatic dependent launch, and cluster launch control. Shape-specific graphs cache encountered widths and depths. A fixed-topology control accepts an updated device configuration without recapture. These controls matter architecturally: the OS must preserve good fused arithmetic and plan reuse even when it selects an execution backend other than persistence. It must not create an artificially weak graph baseline to justify a resident runtime.

a) A process-lifetime resident service.: The architecture also permits a resident executor to accept requests from a host-visible ring, perform sampling and stopping decisions on the device, and publish tokens through an output ring. This is the process-lifetime execution model demonstrated for Llama-3.1-8B by OpenInfer, which also assigns selected CTAs background roles at phase boundaries [1]. Such a service needs request-slot generations, backpressure, cancellation, and a rule for retiring every output consumer. A full ring must stop admissions without blocking token delivery for accepted requests. A cancelled request retires only after its current phase commits and references to its KV are released. Our frontier-level executor provides bounded model execution but does not implement this complete service loop.

D. Three meanings of suspension

GPUOS assigns an explicit suspension mode to a backend. A resident cooperative worker saves progress and stops useful issue while retaining its CTA allocation. A finite sliced backend commits and returns from the kernel, releasing CTA allocation before resumption. A separate-context backend relies on driver-managed switching, whose saved physical execution state is opaque to the application. All can preserve logical task progress, but only the second exposes an application-controlled kernel boundary with resource return.

V. Slack Admission, Dispatch, and Revocation

A. Slack is a budget with a scope

An urgent request has a release time, a completion objective, and a current estimate of its remaining service demand. The architecture distinguishes first-token latency, per-step latency, and whole-request completion: satisfying one does not prove the others. A background admission spends only slack defined at the same boundary as the controller's latency observations. Speculative tokens, variable search width, and prefill/decode transitions require the controller to change its foreground class or invalidate the corresponding calibration.

For the measured sharing experiment, the objective is a step deadline $D_f = 1.06\,Q_{.99}(T_f^{\mathrm{solo}})$. The controller reports both $Q_{.99}(T_f^{\mathrm{co}})/Q_{.99}(T_f^{\mathrm{solo}}) - 1$ and deadline attainment, the fraction of observed steps no longer than D_f. These are different statistics. Passing a p99 ratio test does not establish that every step met its deadline; observing a finite sample with no misses does not establish a deterministic deadline guarantee.

The intended admission problem is to choose background utility subject to foreground and resource constraints:

$$\max_{\mathcal{A},\mathbf{r}} U(\mathcal{A},\mathbf{r}) \quad \text{s.t.} \quad \widehat{T}_f + \widehat{C}_f(\mathcal{A},\mathbf{r}) + G_f \le D_f. \tag{3}$$

Here $\mathcal{A}$ is the admitted task set, $\widehat{C}_f$ includes predicted interference, and G_f reserves revocation and uncertainty margins. Capacity, execution-resource, and protection constraints apply independently. Utility can value completed work, throughput, or application priority. A scalar FLOP count is appropriate only when comparing work whose output and precision make that count meaningful.

This formulation does not imply that a useful global optimizer is already available. The implemented policy uses a measured table for a known workload. Among numerically qualified cooperative configurations whose calibration p99 meets the budget, it chooses the highest measured background

rate. Unknown configurations are not admitted by extrapolation. Selection is frozen before held-out samples are collected. If a selected window fails, the controller revokes background admission for a later window; it cannot repair a deadline already missed. The experiment tests this narrow selection rule, not prediction under arbitrary future workload shifts.

B. The cost model must include the foreground

A useful cost table records an allocation-only baseline in addition to a co-run result. Suppose a foreground loses a units of latency when its worker allocation shrinks, even with no background arithmetic. If a co-run then loses $a + b$, attributing all $a + b$ to memory contention misdiagnoses the problem. Conversely, assigning a separate SM subset does not remove shared cache, DRAM, power, or interconnect effects, so attributing the entire change to lost SMs is also unjustified.

C. Dispatch and safe revocation

The lease lifecycle is *prepared*, *eligible*, *running*, *revoking*, *checkpointed*, and finally *retired*. Preparation verifies tensor views, code versions, resource declarations, and storage availability. Eligibility requires predecessor completion and a valid grant. Revocation prevents new grants, requests a safe point from already-running work, and waits for the specified acknowledgement. Retirement is separate: it also requires that downstream users release the task's state and outputs.

A revocation acknowledgement means that the advertised restart state is committed and that the promised resources have been returned. It must not mean merely that one thread read an urgent flag. A resident worker can only acknowledge its weaker contract: arithmetic issue has stopped, while its CTA allocation remains. A finite-slice backend acknowledges after its CUDA stream confirms completion. The architecture attaches this semantic mode to telemetry so the policy cannot compare unlike acknowledgements as though they offered identical scheduling opportunities.

Let T_{pub}, T_{drain}, and T_{ack} denote control publication, remaining non-preemptible work, and acknowledgement cost. A software revocation estimate is

$$T_{\mathrm{revoke}} \approx T_{\mathrm{pub}} + \max_i T_{\mathrm{drain},i} + T_{\mathrm{ack}}. \tag{4}$$

It becomes a bound only if each constituent is bounded under the admitted interference and the control path is guaranteed execution capacity. A quantum defined as K tiles does not provide that guarantee by itself. Cache misses, context scheduling, and competing traffic can stretch the time of the same instruction sequence. This is why the specification retains a hardware pause requirement even when compiler slicing is available.

D. An IPC implementation of admission

The process-level prototype coordinates one urgent client and a fixed set of background clients with a shared condition variable. A background process acquires the condition lock, verifies that no urgent interval is active, and marks one slice in flight. It releases the lock, launches its finite slice, and waits for stream completion. It then clears its in-flight mark under the same lock and notifies waiters. The foreground sets urgency under that lock and waits until every in-flight mark is clear before submitting its work.

Using one lock for the check and the in-flight transition closes an important race. If a worker first checked an event and only later published its active state, the foreground could observe the old idle state and admit itself just before that worker launched another slice. The foreground would then have a false acknowledgement. The condition-variable protocol gives admission a linearization point. It still depends on trusted workers reporting stream completion honestly; it is not a security boundary against a malicious CUDA client.

E. A worked admission and revocation sequence

Consider an urgent decode client and an adapter-update client whose ready matrix product has been lowered to output tiles and K chunks. This example specifies the proposed protocol; it is not an additional experiment. Before sharing begins, the controller qualifies the foreground geometry, prepares the background executable, allocates its partial accumulators, and resolves all input dependencies. It records a code epoch and a lease generation with the execution cursor. Compilation and first-use initialization occur before the urgent client depends on a bounded revocation interval.

Suppose the foreground announces an eligible sharing interval. The controller looks up a qualified allocation and checkpoint policy and subtracts its revocation reserve from the available slack. A positive remainder permits one background grant. A nonpositive remainder admits no work, even if tensor activity is low. The grant identifies the task, generation, allowed program range, and K extent. The backend issues only that range; the next grant may change its program count without changing which tiles have already committed. This makes resource resizing a change in future scheduling rather than an implicit restart of completed arithmetic.

When urgent work becomes ready, the broker closes admission before waiting for outstanding slices. The finite-slice executor finishes its current launch, commits the next cursor, and acknowledges stream completion. A stale completion from a previous lease generation cannot satisfy this acknowledgement. Only then does the foreground receive the resources promised by this backend. If the background is instead a resident cooperative worker, the acknowledgement has a different meaning: its arithmetic pauses, but its register and shared memory allocation remains occupied. The controller must have admitted that retained footprint in advance.

After the urgent interval, the controller validates the foreground class and lease generation again. An unchanged workload may reuse the qualified policy; a changed context shape or model phase can require a different entry or solo execution. Resumption consumes the last committed accumulator and cursor, not the original inputs alone. Once all output tiles finish, a completion event releases dependent work. Storage is reclaimed only after those consumers retire. This sequence

joins admission, execution, and lifetime management without treating a returned host call as proof of device completion.

F. Failure, cancellation, and ownership

A general OS contract needs a terminal failure state as well as a successful checkpoint state. The specification distinguishes user cancellation, an expired admission lease, a worker that stops responding, and a device error. Cancellation first prevents future grants. It does not retroactively erase stores or DMA operations already issued. An output is published as successful only after its completion condition holds; a cancelled partial output remains private until its in-flight accesses are known to have ended.

The proposed recovery record contains the task identifier, executable epoch, lease generation, last committed cursor, and the ownership of every durable partial buffer. Replaying from that record is valid only if the associated inputs and executable remain available and the task's declared side effects permit replay. Pure tensor stages make this rule tractable. A kernel that updates external state or participates in a collective needs a separate commit protocol; the scheduler cannot manufacture exactly-once effects by relaunching it after a timeout.

A timeout is evidence of missing acknowledgement, not evidence that memory is safe to reuse. If a worker process dies while device work may remain, the controller must obtain a lower-layer completion or teardown guarantee before recycling its allocations. Until then it quarantines the affected lease or reports failure to the foreground. A proposed deadline guarantee must budget this failure behavior separately from ordinary checkpoints. The current trusted IPC prototype checks its local admission lifecycle; it does not implement durable crash recovery, hostile-client containment, or a bounded device-reset service. Those requirements define how the architecture would extend beyond the tested cooperative setting.

G. Progress, fairness, and overload

Foreground protection can starve background work. A controller that pauses backgrounds for every full decode step may be safe for the foreground but economically useless when those steps occupy almost the entire timeline. The architecture therefore reports background progress alongside deadline attainment. A long interval of zero progress should trigger reconsideration of the lease, migration to another device if available, or explicit admission rejection rather than indefinite retention of scarce memory.

VI. Memory and Resource Governance

A. Storage lifetime is independent of execution lifetime

Memory capacity often determines whether another workload can be admitted even when arithmetic is available. A process may hold model weights, request-private KV, shared prefixes, scratch, and checkpoints with very different lifetimes. The proposed allocator names these classes separately. It tracks ownership, consumers, mutability, residency, and reclaim cost, rather than treating all allocated bytes as equally revocable.

An immutable weight image can be retained while its process has no runnable task. A private KV page belongs to a logical request, not to the current position of that request in a compacted batch. A shared prefix requires reference-counted ownership and copy-on-write before divergence. Scratch can be recycled only after the last producer and consumer complete. A partial reduction is durable task state until the task retires, even though the kernel that produced it has returned. These distinctions are essential if execution is to be paused or reorganized without corrupting storage.

B. Capacity leases and eviction

The architectural capacity lease includes retained bytes, an eviction class, and an expected restoration path. Eviction can mean dropping recomputable scratch, writing checkpoints to host memory, preserving an immutable host weight image, or refusing new requests whose KV would exceed the pool. These choices have different costs and correctness requirements. A byte of unused reserved capacity can be cheap to lend; a byte holding the only copy of an active request's state may be expensive to reclaim.

Revocation of memory is a transaction. The controller first prevents new users, then waits for existing device accesses, produces a durable copy if required, updates the residency map, and finally returns the physical allocation. An application-visible pointer cannot remain valid after its backing storage has been reassigned unless the lower layer supplies the corresponding address-translation guarantee. The design therefore separates logical handles from physical mappings and requires versioned residency events for asynchronous consumers.

There is no assumption that paging is always preferable to rejection. Moving state consumes copy engines and host links, may disturb the foreground's memory traffic, and can make subsequent restoration exceed its latency objective. A controller should compare those costs with the remaining useful background work. It should also account for the cost of re-prefill if KV state is discarded. Capacity utilization without restoration cost is an incomplete admission metric.

C. Bandwidth pacing and cache pressure

The implemented resident worker can impose a minimum interval between CTA slice starts. It does not accumulate unused credits into a later burst. This provides a simple logical issue-rate control and prevents a long idle period from authorizing an unbounded foreground-interfering burst. The descriptor records operand and checkpoint bytes, but the control does not enforce a physical DRAM or L2 bandwidth ceiling.

A stronger governor would coordinate budgets across clients and resource domains. Per-worker pacing alone scales the aggregate allowed traffic with the number of workers. Reusing the same per-worker quota after doubling the grid can therefore double the intended issue envelope. The lease specifies whether a budget is per worker, per process, or per device. Global budgets require either a shared credit mechanism or

hardware support; they cannot be inferred from independent local sleeps.

D. Copies, links, and power

Host transfers, peer transfers, collectives, and compute can contend for shared queues or memory bandwidth even when they use different engines. The proposed contract tags a transfer with its direction, source and destination, byte count, deadline class, and whether it is necessary to unblock urgent work. Copy admission must reserve the corresponding memory traffic as well as the copy engine. Otherwise a nominally independent DMA operation can exceed the foreground's memory budget.

Power and frequency are shared state rather than per-task constants. A background workload can alter device clocks and thereby change both foreground and background service rates. The experiments use unlocked clocks and record that limitation; they do not derive a power-aware price model from isolated timing ratios. The proposed governor would need calibrated telemetry and an explicit policy for thermal or board-power constraints. It cannot obtain a per-process power guarantee by adjusting only the number of submitted CTAs.

E. Distributed extensions and tail composition

Although the current scope is one 32-GB GPU, the architecture admits a distributed coordinator above device-local leases. A collective phase must not be admitted independently on ranks whose progress depends on one another. Pausing a rank inside an uncheckpointable collective can strand peers and invalidate a local slack estimate. Collective entry and completion therefore become explicit gang-scheduling boundaries until a lower layer provides a stronger protocol.

For illustration, if independent rank completion times have CDF $F(t)$, the latest of R ranks has CDF $F(t)^R$. A local quantile constraint is not automatically a job-level constraint. Real ranks are correlated by shared inputs, topology, and collective timing, so the independence expression is a model rather than a deployment measurement. The design requires the coordinator to price the observed job tail and to distinguish actual rank measurements from projected maxima. No multi-GPU result is claimed in this paper's single-device experiments.

VII. Prototype Evidence and Evaluation

The evaluation asks whether the architecture's component mechanisms have useful and correctly identified effects. It does not test a completed general-purpose GPU OS. We separate four evidence scopes: qualified full-model execution, resource-sharing microbenchmarks, compiler correctness, and a cross-process context-switch diagnostic. They have different timing boundaries and cannot be combined into a single system speedup.

A. Platform and qualification

The local experiments use one NVIDIA RTX 5090 with 170 SMs, 32 GB of memory, compute capability 12.0, CUDA 13.1, and driver 610.57.04. The model environment uses PyTorch 2.14.1+cu130. CPU and GPU clocks are not locked. The experiment records CPU affinity and checks competing compute processes; residual host, clock, and thermal variation remains. CUDA events measure device intervals, while host monotonic clocks measure request or process-visible latency.

Evidence	Boundary and replication
Complete requests	Fresh prefill through final D2H/text; 3 processes, 5 requests per cell.
Search replay	Ready ancestor through fixed native search trace, with scoring timed; 3 process ratios.
Slack sharing	256 calibration and 512 held-out steps per cell, paired adjacent solo; 3 processes.
Context diagnostic	20 paired same-/different-process trials; one warmed pair of CUDA processes.
Compiler checks	Seven graph/lifecycle tests plus separate generated-kernel sanitizer checks.

TABLE II
DIFFERENT EVIDENCE SCOPES. A PROCESS RANGE IS NOT A CONFIDENCE INTERVAL, AND AN OPERATOR RESULT IS NOT A COMPLETE SERVING MEASUREMENT.

The Qwen3-0.6B checkpoint is revision `c1899de`. The custom fused path must satisfy finite full-vocabulary logits, NRMSE at most 0.01, and cosine similarity at least 0.999 at every tested frontier. Final free-search paths are checked independently. A racecheck audit found a missing warp barrier between reading and overwriting the online-softmax denominator. The retained Figure 1 and Figure 2 results use the repaired, rebuilt path and fresh measurements. Earlier unqualified timing records are not substituted for the corrected evidence.

Complete-request timings include native prefill, root-state publication, all model frontiers, branch scoring, private-history copies, final decoding, and synchronization in one outer interval. Weights, tokenized inputs, workspace, and reusable graphs are resident. We do not add stage medians to construct request latency. Figures retain medians and ranges across independent processes. A small range in such a plot is not an estimate of the probability that a future service request meets a deadline.

B. Complete-request benefit

The FuseOS executor retains an application-level benefit after the attention repair. For 32 search steps, the complete request takes 117.9 ms, compared with 143.3 ms for native shape-cached execution (1.215×). For 64 steps, the corresponding times are 225.0 ms and 274.9 ms (1.222×). Figure 1 includes the native GPU-length graph, which keeps graph count bounded by frontier width rather than sequence depth. Thus the comparison is not against repeated Python launches or compulsory graph reconstruction.

All 28 decoder layers and the vocabulary head execute in these intervals. The searches expand 128 and 251 branch-tokens respectively; the latter changes the live frontier between three and four branches. Both use model outputs for par-

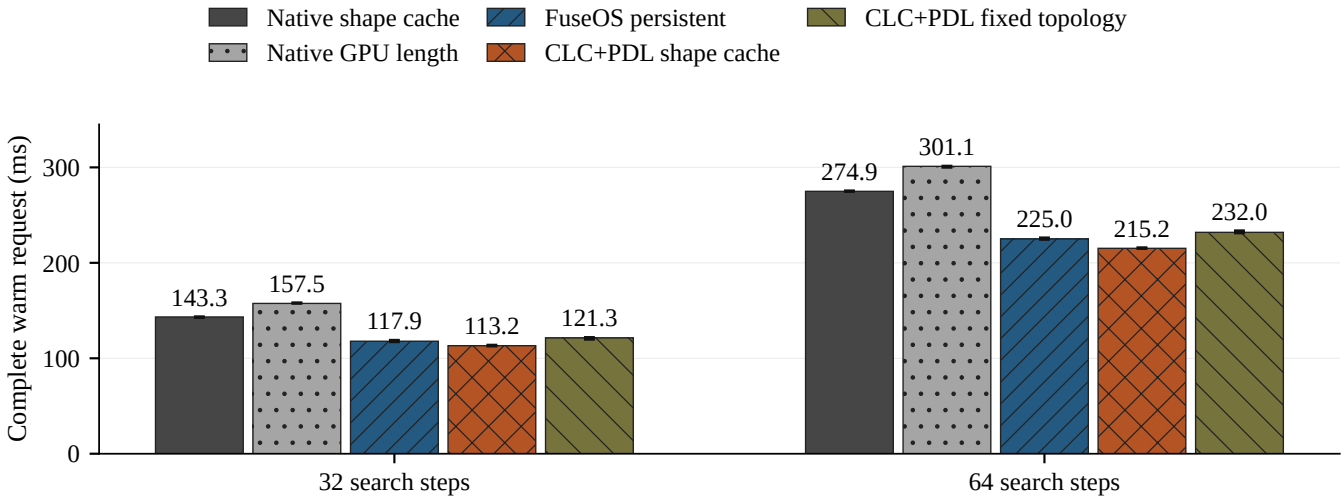


Fig. 1. **Complete Qwen3-0.6B requests after synchronization repair.** Each request freshly prefills a 128-token ancestor and performs output-driven search with at most four branches. Bars start at zero; whiskers span three process medians (five requests per process). Shape-cache controls reuse all encountered graphs; the fixed-topology control updates configuration without recapture. All five methods produce the same final paths for each horizon. The benefit over native execution includes the fused implementation as a whole. Cached CLC+PDL remains a stronger executor than persistence here.

ent/token selection, and the corrected custom paths reproduce the final native candidate paths. The maximum request-gate NRMSE is 0.007762, below the unchanged threshold.

a) What the positive result establishes.: The complete dynamic executor is useful for this narrow-frontier workload. The result does not isolate persistence from fused arithmetic. With the same fused handlers, shape-cached CLC+PDL completes the two requests in 113.2 ms and 215.2 ms, faster than the persistent executor. The fixed-topology CLC+PDL control takes 121.3 ms and 232.0 ms. This comparison distinguishes the cost of the execution protocol from the benefit of the fused implementation. We retain the full run ranges, including prefill variation, instead of presenting the small fixed-topology differences as a universal persistent advantage.

C. Dependence on workload geometry

We next hold the search trajectory constant to remove work differences from the backend comparison. Three configurations vary ancestor length P, maximum frontier width B, and horizon H: $(128, 4, 32)$, $(512, 16, 16)$, and $(1024, 8, 32)$. Each uses a native-generated, output-dependent trace. All qualified executors perform the same selected histories and execute the selection policy during timing.

Figure 2 shows that the narrow-frontier benefit does not extend uniformly to wider or longer-prefix cases. Persistence improves on the matched tile-grid and tile+PDL implementations, but native cached execution becomes competitive or faster as workload geometry changes. Shape-cached CLC+PDL provides another strong control. Since prefix, width, horizon, and trajectory all change between cases, this figure defines an observed applicability boundary rather than identifying a single causal shape parameter. The maximum post-fix replay NRMSE is 0.008308; timing includes no gate-failing configuration.

D. Mechanism and larger-model controls

Matched attention controls distinguish placement and task partitioning from execution fusion. With identical 512-key jobs, replacing round-robin placement with length-aware placement gives a geometric-mean 1.234× speedup and a maximum 1.925×. Changing an already sorted ordinary grid from four partitions to finer 128-key jobs gives 1.843× geometrically, up to 3.988×. Both exploit information known before dispatch. They must not be presented as evidence that a device discovered new dynamic work more effectively.

The matched CLC phase-fusion control instead retains the same compiled entry, task geometry, resources, and global exchange buffers, switching between staged and last-producer reduction. Its geometric-mean increment is 0.994×. The change removes a launch but adds completion and ordering work. Separate cluster-transport controls likewise do not establish a large universal benefit from moving state across SMs. Their implication for the OS design is to preserve good allocation and dataflow plans before choosing a more elaborate execution mechanism.

The Qwen3.5-35B-A3B port provides another boundary. It covers routed MoE work for microbatches of one to eight tokens and the DeltaNet decode path, while other operations remain native. Its default fused path does not beat native eager execution across the six evaluated workloads. Separating selected stages reduces two request times from 721.11 to 682.92 ms and from 554.12 to 547.39 ms. More fusion is therefore not an optimization objective by itself. This separate

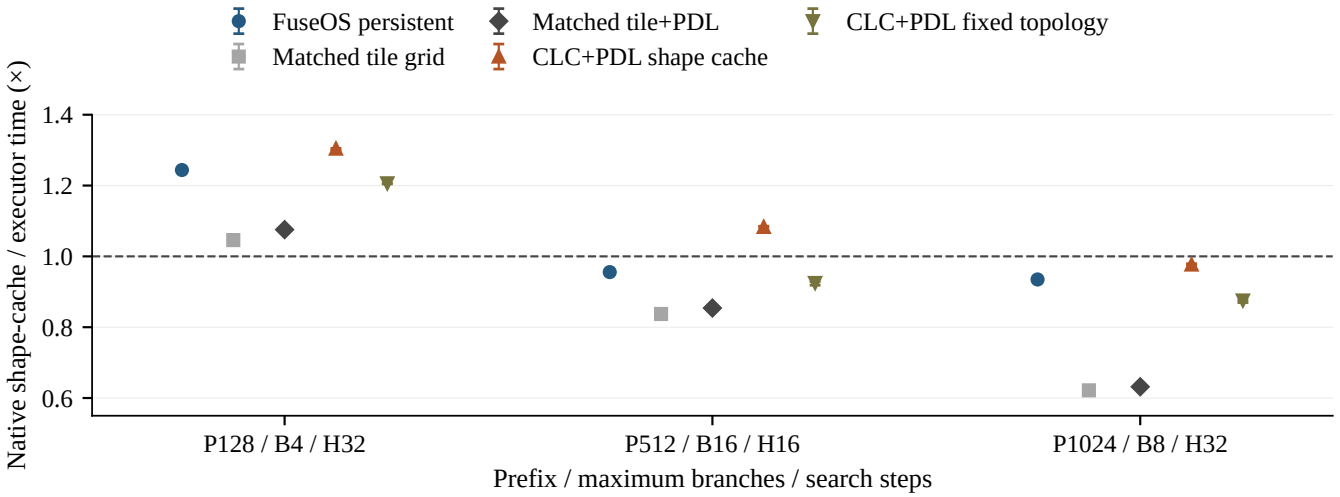


Fig. 2. **Full-model equal-work search replay.** Native output-dependent parent/token decisions are replayed for every backend while scoring still executes inside timing. The interval starts from a ready prefilled ancestor; it is not the complete-request interval in Figure 1. Higher is faster; 1× is native shape-cached execution. Every point includes all layers, KV updates, host policy, and synchronization. Points and whiskers are the median and range of three round-indexed ratios.

mixed-Q4 comparison does not establish BF16 accuracy or a port of every model operator to the new FX compiler.

E. Slack sharing and its application boundary

The resident-worker experiment runs independent 16×16×4096 BF16 GEMMs with FP32 accumulators, four warps per CTA. After a configurable number q of K tiles, each warp saves its accumulator and progress. The foreground publishes an urgency flag on its own stream. Background workers observe it at checkpoints but retain their allocations while paused. Non-yielding long- and short-quantum controls isolate the effect of responding to urgency from merely changing the amount of work in a quantum.

The streaming foreground contains eight independent BF16 GEMVs with $M = 2048$ and $K = 8192$, reading 256 MiB of weights per step. Zero- and 100-μs device gaps provide distinct opportunities for background work. This is not a transformer or a Kimi shard. The model foreground executes all Qwen3-0.6B layers and the vocabulary head using native cached graphs, with a 128-token prefix, up to four branches, and a 32-step native trace. Unlike Figure 2, this sharing replay excludes search scoring; unlike Figure 1, it excludes prefill. That distinction is preserved in the results.

Each configuration has an adjacent standalone reference, randomized order, 256 calibration steps, and 512 held-out steps in each of three fresh processes. CUDA-event step latency includes the urgency markers and graph execution but excludes the intentional gap. Background rate counts executed K tiles over the host start-through-stop window, including startup, gaps, submission, and the last slice drain. The artifact retains 154,368 foreground timings including calibration and references.

With 100-μs gaps, the selected 64-worker, $q = 64$ configuration provides 1.167 TFLOP/s for 2.50% p99 cost. Shorter $q = 4$ checkpoints reduce cost to 0.62% but provide 0.741 TFLOP/s. The long non-yielding control produces 4.359 TFLOP/s at 23.68% cost and misses the budget. Without an intentional gap, the selected configuration provides 0.374 TFLOP/s at 3.48% cost; launch and event boundaries still exist in that case.

The table's normalized rate uses the fastest measured configuration of the same custom GEMM as its denominator. It is not a fraction of hardware peak and cannot be compared numerically with another implementation's "GPU worth" without matching its work and denominator. All nine frozen calibration selections choose the 64-worker, $q = 64$ policy, and all nine pass held-out validation. Those observations do not establish generalization to a new model, a new load trace, or a changed driver.

The full-model boundary is more important than the positive proxy point. Whole-step suspension provides only 0.039 TFLOP/s, or 0.342% of the custom solo rate. The non-yielding long-quantum control instead provides 8.779 TFLOP/s while satisfying the observed p99 budget. It lacks a prompt-yield contract, but it is a stronger performance control for that workload. The result argues for phase-sensitive admission rather than protecting an entire model step whenever a smaller interval would suffice. Negative measured latency costs are retained under unlocked clocks and are not credited as a scheduler-induced speedup.

F. Checkpoint correctness and pause observations

The handwritten worker checks all 256 outputs per warp and saved partial accumulators against independent CPU

Background policy	p50 cost (%)	p99 cost (%) [range]	TFLOP/s	Solo rate (%)	Passes
Streaming GEMV proxy, 100-μs gaps					
No yield, q4096, w64	+22.06	+23.68 [+23.64, +25.87]	4.359	38.252	0/3
No yield, q4, w64	+18.70	+20.91 [+20.77, +23.90]	2.852	24.964	0/3
Yield, q4, w64	+0.58	+0.62 [+0.58, +0.99]	0.741	6.507	3/3
Yield, q64, w64	+2.34	+2.50 [+2.32, +2.61]	1.167	10.213	3/3
Paced, q4, w32	+1.17	+1.16 [+1.15, +1.30]	0.055	0.484	3/3
Qwen3-0.6B native cached-graph decode replay					
No yield, q4096, w64	-1.80	-1.57 [-1.67, -1.46]	8.779	76.999	3/3
No yield, q4, w64	+4.44	+3.92 [+1.36, +4.24]	3.263	28.537	3/3
Yield, q4, w64	-3.40	-4.42 [-4.47, -3.14]	0.011	0.094	3/3
Yield, q64, w64	-3.25	-3.04 [-4.44, -2.48]	0.039	0.342	3/3
Paced, q4, w32	-1.58	+0.37 [-1.51, +0.42]	0.001	0.012	3/3

TABLE III

CO-RUNNING ON RTX 5090. VALUES ARE MEDIANS OF THREE PAIRED PROCESS RATIOS; BRACKETS ARE THEIR RANGES. PASSES COUNT HELD-OUT PROCESSES WITHIN THE 6% P99 BUDGET. SOLO RATE IS NORMALIZED TO THE FASTEST MEASURED CUSTOM GEMM, NOT HARDWARE PEAK. HERE q IS K TILES PER WARP BETWEEN CHECKPOINTS AND w IS BACKGROUND CTAS, NOT RESERVED SMS.

arithmetic on binary-exact operands. Resume tests include a quantum that leaves a partial GEMM. The streaming outputs remain bitwise stable and selected rows pass an independent FP32 dot-product check. The final replay's model logits match the native reference bitwise at every step. Bounded runs of the same worker entry pass memcheck, racecheck, and synccheck with kernel filters. An initial live-worker sanitizer attempt stalled and was terminated, so the bounded checks are not described as certification of all concurrent execution.

A separate device probe measures urgency publication until all 64 workers acknowledge suspension. Its p99 is 10.69, 21.85, and 738.68 μs for $q = 4$, 64, and 4096 respectively. The diagnostic polling kernel perturbs execution and excludes CPU dispatch; its timings do not enter throughput aggregates. Non-cooperative workers time out rather than acknowledging a yield. The large change with quantum illustrates why block or slice lifetime is a first-class admission cost.

The FX compiler has additional checks for broadcast/fan-out graphs, odd matrix dimensions, FP32/FP16/BF16 arithmetic, linear/GELU compositions, reductions and normalization, views, the `torch.compile` entry point, and rejection/lifecycle behavior. Tests vary the program-count grant during a partially completed reduction. Generated matrix, reduction, and pointwise kernels pass separate bounded sanitizer checks. These establish correctness for the supported cases; they do not imply coverage of arbitrary Python, arbitrary CUDA kernels, training, or the complete Qwen graph.

G. Existing hardware context switching

The context-switch diagnostic deliberately uses non-cooperative work. A background kernel launches one CTA on every SM, with more than half the per-SM shared memory reserved so that only one such CTA can reside per SM. Each CTA executes a 200-ms loop without an urgency poll. Its arithmetic state and shared-memory sentinel are verified after completion. The foreground probe reserves the same amount of shared memory, preventing it from being explained by spare same-context residency.

Probe placement	p50 (ms)	p99 (ms)	Full overlap
Same context, high priority	199.910	199.930	0/20
Different CUDA process	2.000	2.046	20/20

TABLE IV

WARMED SUBMISSION-TO-COMPLETION LATENCY FOR A 10-μS PROBE. FULL OVERLAP MEANS ALL 170 BACKGROUND CTA LIFETIMES ENCLOSE THE COMPLETE PROBE ACCORDING TO DEVICE TIMESTAMPS. ONE PROCESS PAIR, 20 TRIALS PER MODE.

The device advertises compute-preemption support. GPU timestamps and SM identifiers confirm that, in every different-process trial, the probe runs while all 170 background CTA lifetimes remain open. The corresponding register computation and shared-memory state checks pass. In the same context, higher stream priority does not interrupt the full-residency background. These observations are consistent with driver-managed hardware context preemption [7]; they do not expose its internal save format or measure the context-save operation in isolation. The reported latency includes driver scheduling and submission overhead.

The process harness also runs one native Qwen foreground with one or three compiler-generated GEMM background processes on this same device. Separate contexts and a private MPS service are distinct configurations. Client PIDs and MPS server attachment are recorded, and interrupted partial background work is completed and checked after timing. The broader repeated process sweep was paused during this paper's design-focused revision; partial rounds are retained but are not used to assert an aggregate service-level speedup. Single-device multiprocessing is not presented as multi-GPU reproduction.

VIII. HARDWARE AND MODEL CO-DESIGN

The component results support the source paper's premise that software scheduling and kernel design cannot by themselves manufacture every missing device guarantee. We organize the hardware interface around three primary requests and supporting observability and control services, following

the agenda of Nourai and Bastani [1]. These are requirements of the proposed architecture, not capabilities claimed for the prototype.

A. S1: bounded pause with state-preserving resume

The desired primitive accepts a task or lease identifier, requests a pause, and acknowledges within a specified time bound while preserving restart state. Its contract must say whether it returns CTA resources, whether memory transactions are drained, how barriers are preserved, and what completion guarantees remain valid. A flag that stops useful issue without returning allocation does not satisfy the same interface.

The source agenda's 5–20-μs target is substantially stronger than observing that an independent CUDA process eventually runs. Our approximately 2-ms different-process probe latency is evidence that non-cooperative work need not monopolize this device, but it is neither an application-selected deadline nor a microsecond pause guarantee. Compiler-generated kernel boundaries provide another usable control point, at the cost of submission, checkpoint traffic, and lost locality. Both remain relevant even if future hardware reduces one part of the revocation cost.

B. S2: priority and rate control in shared memory paths

SM separation alone cannot prevent L2 or DRAM interference. The desired memory interface tags transactions by task class and supports policies for urgent queueing, background rate limits, and observable enforcement. A meaningful guarantee must distinguish bandwidth from latency and must define the granularity at which priorities can change. Cache partitioning alone does not provide a bound on memory-controller queueing delay.

The controller must also avoid starving traffic that is required for forward progress. A background checkpoint or an outstanding write can be necessary before resources become reclaimable. Treating all such traffic as indefinitely deferrable can prevent the urgent lease from acquiring those resources. The proposed tagging model includes dependency-critical completion traffic and a bounded drain class in addition to foreground and background payloads. The current CTA issue pacing does not implement this memory-controller policy.

C. S3: enforceable per-kernel partitions

The desired partition contract states which execution resources a kernel can use and is honored consistently across ordinary launch, recorded graphs, and changes of model phase. Static partitions can still be useful, but rapidly alternating phases require either inexpensive partition changes or a scheduler that selects appropriately provisioned queues. Merely accepting a partition parameter in an API is not evidence that the physical execution honors it.

A phase-dependent controller must account for its own transition cost. Switching streams, queues, or contexts can introduce bubbles, synchronization, and cache disruption. A promising single-layer result can consequently fail when every layer pays the transition. The proposed planner evaluates complete steps and includes partition transitions in the cost model. It does not add the best isolated-phase throughput values and call the sum a feasible model schedule.

D. Supporting services and observability

Priority inside an SM would complement, rather than replace, admission and memory control. Urgent instructions sharing a resident allocation can still be delayed by dependencies or transactions already issued by other work. Fast device-wide signals and queues are useful when they have explicit memory-ordering and progress guarantees. Accessible counters should separate residency, arithmetic activity, traffic, and queueing instead of exposing one ambiguous utilization number. Priority and pacing for copy engines, links, and power complete the resource-vector interface.

E. Models that produce predictable sharing opportunities

Model architecture can make the scheduling problem easier without replacing the OS contract. Balanced expert routing and similar expert sizes can reduce phase variability. Linear-attention components can avoid some context-length growth in the recurrent state update. Speculative or multi-token processing can change the arithmetic reuse of a step. Each choice also changes numerical semantics, latency, or memory use and must be evaluated with the actual model. We do not infer a service speedup from regularity alone.

IX. Related Work and Scope

The resource-vector and slack-budget framing is directly inherited from *The GPU Needs an Operating System*; its large-model measurements and hardware proposals motivate this specification [1]. Our emphasis is the executable meaning of safe points, resource return, publication, and admission, together with the boundaries visible in local prototypes. Reframing those ideas as an API is not a claim of having invented GPU sharing or having implemented the entire proposed stack.

FX provides the program-capture substrate and Triton the tiled code-generation substrate [5], [6]. The present compiler adds a restricted checkpointed execution plan; it is not a replacement for either general compiler framework. MPK combines a compiler and runtime for mega-kernelizing tensor programs [8]. Meld couples dynamic work placement, cross-SM cooperation, and just-in-time reduction [9]. Our matched controls show why allocation, fusion, and transport must be separated when attributing a speedup. Moving intermediate data or claiming an unstarted CTA is not migration of a live program counter.

CUDA context scheduling, stream priorities, green contexts, and MPS are lower-layer mechanisms with distinct contracts [2]–[4], [7]. The architecture uses them where their documented guarantees match a lease. It does not equate MPS clients with independent devices or a software memory quota with hardware fault isolation. PagedAttention demonstrates the value of treating KV allocation and sharing as first-class serving concerns [10]; scheduling the compute that accesses those pages requires additional lifetime and revocation rules.

X. Limitations and Research Questions

The complete object model remains an architectural specification. The current artifact contains bounded fused inference, a restricted graph compiler, cooperative workers, an IPC admission prototype, and diagnostics. It lacks an integrated multi-tenant service, arbitrary-kernel transformation, a learned interference predictor, a general memory-lease manager across protected contexts, and the proposed hardware QoS controls. Correctness of supported graphs and component numerical checks is not a proof of whole-system isolation, fault recovery, or deadline satisfaction.

The measurements use one consumer Blackwell GPU with unlocked clocks. They neither reproduce the source paper's RTX PRO 6000 model serving nor establish transfer to AMD, TPU, SRAM-centric accelerators, or multi-node deployments. Roofline reasoning can motivate those investigations, but partition granularity, memory-system behavior, driver scheduling, and communication must be remeasured on each target. In particular, the observed context-switch diagnostic cannot specify a worst-case scheduling bound.

XI. Conclusion

GPU sharing requires more than placing additional kernels beside a model. It requires a contract for what work can stop, what state survives, what resources are actually returned, and how much foreground slack admission is allowed to spend. GPUOS specifies that contract across processes, compiler slices, resource leases, and hardware services. The component evidence supports useful bounded execution and selected sharing opportunities, while exposing costs that simpler utilization arguments miss. The architectural goal is a general GPU operating system; the present contribution is a specified path toward that goal with explicit implementation and evidence boundaries.

a) Artifact and drafting disclosure.: The repository includes raw timings, hashes, numerical checks, and build instructions. Both figures retain qualified data; incomplete sweep rounds are excluded from aggregate claims. AI-assisted tools supported code, analysis, and prose. Results derive from retained measurement records.

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